

Pairwise Phase-Difference Coupling Preserves Fine-Grained Pattern Discrimination in Associative Memory

A Progressive Ablation Study

Labinot Marku, MD

Department of Neurosurgery, KRH Klinikum Nordstadt Hannover, Germany

labinot.marku@krh.de

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Code: <https://github.com/labinot-marku/CONN-Associative-Memory/tree/main/similarity-discrimination>

Abstract

We report a progressive ablation study investigating whether phase-coupled oscillatory networks discriminate highly similar patterns that binary and continuous Hopfield networks cannot. Using families of correlated patterns (70–95% similarity), we measure each system's ability to identify the specific family member from a noisy cue. Through eleven sequential experiments — each designed to address the previous experiment's most important criticism — we isolate the mechanism responsible for the discrimination advantage.

The main findings: (1) A minimal phase-coupled oscillatory network discriminates family members at 92.5% accuracy versus 25.8% for both binary and continuous Hopfield networks under matched noise and evaluation conditions ($p < 0.0001$, 20 seeds). (2) The effect does not depend on amplitude dynamics or the coherence prior — bare phase coupling is sufficient. (3) The effect is robust from 70% to 95% pattern similarity. (4) Continuous Hopfield (tanh activation) performs identically to binary Hopfield, ruling out continuity as the explanation. (5) The critical ingredient is pairwise phase-difference coupling: any tested function of $(\phi_j - \phi_i)$ discriminates (sin, linear, cosine all ~93%), while field-based coupling on the same circular manifold fails completely (25.8%). (6) The phase system preserves 1.764 bits of member-specific mutual information (88% of maximum) versus 0.049 bits for Hopfield ($p = 0.0003$), despite Hopfield achieving higher exact recall accuracy (87.7% vs 74.7%). (7) Discrimination improves with network size under fixed family structure (reaching 100% at $N \geq 192$, though this may partly reflect decreasing effective load), and survives pattern loads up to $\alpha = 0.65$. (8) At $N=1024$ with only 25% sparse connectivity, pairwise models maintain perfect discrimination while field models collapse to exact Bernoulli chance ($25.0\% \pm 43.3\%$).

The mechanism: pairwise phase-difference coupling preserves fine-grained angular relationships between oscillators that global field-based coupling (whether binary sign or continuous tanh) destroys through averaging. Phase-coupled dynamics preserve substantially more member-specific information than binary and continuous Hopfield baselines, despite lower exact-recall accuracy, revealing a separation between identity preservation and bit-level reconstruction. All code is publicly available and all results are deterministically reproducible across independent computational environments.

1. Introduction

Associative memory networks store patterns as attractors in a dynamical energy landscape. The classical Hopfield network (Hopfield, 1982) stores binary patterns via Hebbian learning and

retrieves them through convergence to the nearest attractor from a noisy cue. The network's capacity limit is $\alpha_c \approx 0.138$, meaning the network can store approximately $0.138N$ patterns before retrieval degrades (Amit, Gutfreund & Sompolinsky, 1987).

A fundamental limitation of binary attractor networks, less studied than capacity, is their behavior under pattern correlation. When stored patterns are highly similar — sharing 80–95% of their components — their attractor basins overlap substantially. The network converges to a “prototype” attractor representing the average of the similar patterns rather than the specific pattern that generated the cue. Individual family members become indistinguishable.

This limitation is well-known in the Hopfield literature (Amit, 1989; Hertz, Krogh & Palmer, 1991) and is a consequence of the binary representation: the Hebbian weight matrix $W = (1/N) \sum \xi_\mu \xi_\mu^T$ treats correlated patterns as nearly identical because their outer products are nearly identical. The weight matrix cannot encode the fine-grained differences that distinguish family members.

Oscillatory neural networks (Kuramoto, 1984; Aonishi, 1998; Cook, 1989; Hoppensteadt & Izhikevich, 1999) represent each neuron as a phase oscillator $z_j = A_j \exp(i\phi_j)$, where retrieval occurs through phase synchronization rather than binary convergence. The Coherent Oscillatory Neural Network (CONN; Marku, 2026) demonstrated that phase-amplitude dynamics with a coherence prior achieve higher operational capacity ($\alpha \approx 0.19$) than classical Hopfield under noisy retrieval conditions.

This manuscript does not address capacity. It addresses a different and potentially more fundamental question: does the phase representation enable discrimination between similar patterns that binary representations cannot achieve? We report eleven progressive experiments, each designed to answer the most important objection raised by the previous experiment, culminating in a precise identification of the mechanism responsible.

2. Methods

2.1 Phase-Coupled Oscillatory Network

Each neuron j is a phase oscillator with state $\phi_j \in [0, 2\pi)$. Patterns are stored as binary phase configurations $\phi_j \in \{0, \pi\}$, corresponding to the binary encoding $+1 \rightarrow 0, -1 \rightarrow \pi$ (equivalently, $\cos(\phi_j) \in \{+1, -1\}$).

The Hebbian weight matrix is constructed from the cosine encoding:

$$J_{ij} = (1/N) \sum_\mu \cos(\phi_i^\mu) \cos(\phi_j^\mu), \quad J_{ii} = 0$$

The full CONN dynamics include amplitude A_j , coherence prior $\lambda \sin(2\phi)$, and annealed coherence scheduling. For the ablation study, we progressively remove these components. The minimal surviving system is bare pairwise phase coupling:

$$d\phi_i/dt = \eta \sum_j J_{ij} f(\phi_j - \phi_i)$$

where f is a function of the pairwise phase difference (sin, linear circular difference, or cosine in our experiments), and $\eta = 0.005$ is the integration step size. Integration proceeds for 150 steps.

2.2 Hopfield Baselines

Binary Hopfield: $W_{ij} = (1/N) \sum_\mu \xi_i^\mu \xi_j^\mu$ with synchronous $\text{sign}()$ updates until convergence (max 100 steps).

Continuous Hopfield: $dx/dt = -x + \tanh(\beta W x)$ with $\beta = 2.0$, $dt = 0.1$, 150 steps. Output binarized via $\text{sign}()$.

2.3 Pattern Families

A family of correlated patterns is generated by creating a random binary prototype $\xi \in \{-1, +1\}^N$, then producing $n_members$ variants by flipping $(1 - \text{similarity}) \times N$ randomly selected bits in each variant. Family members share a fraction ‘similarity’ of their components and differ in the remainder. Unless otherwise stated: $N = 80$, 3 families $\times$ 4 members, 5 unrelated patterns, 17 total ($\alpha = 0.212$).

2.4 Evaluation Metric

For each retrieval, the output is binarized and compared against all family members. The retrieved member is identified as the one with highest overlap. Discrimination accuracy is the fraction of retrievals where the correct member (the one that generated the cue) is identified. Random chance for 4 members is 25%.

This metric tests a more stringent property than standard recall accuracy. It asks not “did you get approximately the right pattern?” but “can you distinguish which specific variant among 85%-similar alternatives was the input?”

2.5 Noise Model

From Experiment 5 onwards, all systems receive the same binary-corrupted cue: each bit is flipped with probability $p_noise = 0.25$. For phase systems, this cue is converted to phases plus minimal Gaussian jitter ($\sigma = 0.15$) to activate dynamics. For continuous Hopfield, the binary cue is converted to floats plus the same jitter magnitude. This ensures all systems start from equivalent information.

3. Experimental Progression

Each experiment was designed to address the most important criticism identified by the multi-AI review process (Claude, ChatGPT, Gemini, Grok) applied to the previous experiment’s results.

3.1 Experiment 1: Initial Discovery

Question: Does CONN discriminate family members that Hopfield cannot?

Setup: $N=80$, 3 families $\times$ 4 members (85% similar), 5 unrelated, 25% noise, 15 seeds. Full CONN dynamics (amplitude + coherence prior). Hopfield baseline.

Metric	CONN	Hopfield	Δ
Exact target accuracy	87.7%	88.1%	−0.4%
Correct member identified	99.4%	25.6%	+73.9%
Right family (any >80%)	98.3%	97.2%	+1.1%

Both systems achieve similar bit-level accuracy (~88%), but CONN identifies the correct family member 99.4% of the time while Hopfield operates at chance (25.6%). Wilcoxon $p < 0.0001$.

Criticism raised: Is the effect from amplitude? From the coherence prior? From the noise model? Single comparison against crude baseline.

3.2 Experiment 2: Component Ablation

Question: Which CONN component drives the discrimination?

Configuration	Discrimination	Hopfield	Δ
Full CONN	99.4%	25.6%	+73.9%
No amplitude	99.4%	25.6%	+73.9%
No coherence prior	99.4%	25.6%	+73.9%
No amplitude + no coherence	98.9%	25.6%	+73.3%

Neither amplitude dynamics nor the coherence prior contributes to family discrimination. The effect survives removal of both CONN-specific innovations. The minimal system — bare sinusoidal phase coupling $d\phi_i = \eta \sum_j J_{ij} \sin(\phi_j - \phi_i)$ — is sufficient.

Criticism raised: What about different similarity levels? Computational budget asymmetry?

3.3 Experiment 3: Similarity Sweep

Question: At what similarity level does discrimination fail?

Similarity	Phase-coupled	Hopfield	Δ
70%	100.0%	32.2%	+67.8%
80%	100.0%	26.7%	+73.3%
85%	99.4%	25.6%	+73.9%
90%	98.9%	23.9%	+75.0%
95%	97.8%	25.6%	+72.2%

The effect is robust across all tested similarity levels, with graceful degradation from 100% at 70% to 97.8% at 95%. At 95% similarity, family members differ in only 4 of 80 neurons, yet the phase system still discriminates. Hopfield remains at chance throughout.

Criticism raised: Hopfield gets fewer retrieval attempts.

3.4 Experiment 4: Computational Budget Control

Question: Does giving Hopfield more attempts close the gap?

Hopfield attempts	Phase-coupled	Hopfield	Δ
1	99.4%	25.6%	+73.9%
3	99.4%	25.0%	+74.4%
5	99.4%	25.0%	+74.4%
10	99.4%	25.0%	+74.4%

Ten independent Hopfield retrieval attempts with different noise realizations do not improve discrimination. The information to distinguish family members does not exist in Hopfield's binary weight matrix. More computation cannot extract what is not represented.

Criticism raised: Noise model and evaluation are asymmetric between systems. Need continuous Hopfield baseline.

3.5 Experiment 5: Matched Conditions + Continuous Hopfield

Question: Is the effect from phase dynamics specifically, or from continuity generally?

All systems receive the same binary-corrupted cue. All outputs are binarized before evaluation. Continuous Hopfield (tanh activation) added as baseline.

System	Discrimination	±Std
Phase-coupled (sin)	92.5%	±7.4%
Continuous Hopfield (tanh)	25.8%	±3.6%
Binary Hopfield (sign)	25.8%	±3.6%

Continuous Hopfield performs identically to binary Hopfield (25.8% vs 25.8%, $p = 0.50$). Both are at chance. The tanh activation, which makes neurons continuous-valued, provides zero discrimination advantage. Phase coupling vs binary: $p < 0.0001$. Phase coupling vs continuous Hopfield: $p < 0.0001$.

The effect dropped from 99.4% to 92.5% after matching the noise model, confirming that the original noise asymmetry contributed approximately 7 percentage points. The remaining 66.7% gap under matched conditions is not a methodological artifact.

Criticism raised: Is it the sinusoidal function, the circular topology, or both? Need decomposition.

3.6 Experiment 6: Phase System Decomposition

Question: Which specific ingredient of the phase dynamics is responsible?

Five coupling models tested with matched noise and evaluation:

Model	Coupling	Discrimination	±Std
A. Sinusoidal	$\sin(\varphi_j - \varphi_i)$	93.3%	±5.7%
B. Linear circular	$(\varphi_j - \varphi_i) \bmod 2\pi$	93.3%	±5.7%
C. Cosine	$\cos(\varphi_j - \varphi_i)$	92.5%	±5.8%
D. tanh on circle	$\tanh(J @ \varphi)$	25.8%	±9.5%
E. Binary Hopfield	$\text{sign}(Wx)$	25.0%	±2.6%

Models A, B, C — all using pairwise phase-difference coupling — perform identically (~93%). Models D and E — using global field-based coupling — both fail (25–26%). The coupling function does not matter ($\sin \approx \text{linear} \approx \cos$). The coupling structure does: pairwise difference versus global field.

Model D is critical: it lives on the same circular manifold S^1 as Models A–C but uses field-based coupling $\tanh(\sum J_{ij} \varphi_j)$ instead of pairwise differences. It fails completely. The circular topology alone is not sufficient. Pairwise phase-difference coupling is essential.

3.7 Similarity Sweep Across All Five Models

Similarity	sin	linear	cos	tanh-circle	binary
70%	99.6%	99.6%	99.6%	19.2%	38.3%
80%	97.1%	97.1%	97.1%	27.1%	25.4%
85%	93.3%	93.3%	92.5%	25.8%	25.0%
90%	85.0%	85.0%	84.2%	22.1%	25.4%
95%	70.8%	70.8%	70.0%	26.7%	25.0%

All three pairwise-difference models track together perfectly across the full similarity range. Both field-based models remain at chance. The separation is complete and consistent.

3.8 Experiment 7: Mutual Information Analysis

Question: Does the phase system preserve more member-specific information, not just classify better?

We compute $I(\text{target member; identified member})$ in bits for all three systems. Maximum possible $MI = 2.0$ bits (4 family members). $N=80$, 85% similarity, 15 seeds, matched conditions.

Metric	Phase	Cont Hop	Bin Hop
Discrimination	92.8%	25.6%	26.1%
Exact recall accuracy	74.7%	87.3%	87.7%
MI (bits, max 2.00)	1.764	0.074	0.049
MI / max MI	88.2%	3.7%	2.5%

Phase MI vs Binary MI: $\Delta = +1.715$ bits, $p = 0.0003$. Phase MI vs Continuous MI: $\Delta = +1.691$ bits, $p = 0.0003$. The phase system preserves 88% of maximum member-specific information. Both Hopfield variants preserve approximately zero.

The exact recall inversion.

Hopfield achieves higher exact recall (87.7%) than the phase system (74.7%). Yet Hopfield preserves zero member-specific information while the phase system preserves 88%. This reveals that the two systems optimize different objectives: Hopfield converges to the family prototype (high bit accuracy against any member, zero discrimination), while the phase system converges to the specific member (lower bit accuracy but high structural fidelity). Standard bit-level recall accuracy is blind to this distinction.

3.9 Experiment 8: Scaling

Question: Does the effect survive at larger N ?

N	Ph disc	Bn disc	Ph MI	Bn MI
80	91.7%	26.7%	1.747	0.049
128	99.4%	25.0%	1.982	0.000
192	100.0%	25.0%	2.000	0.000
256	100.0%	25.0%	2.000	0.000

The effect strengthens with network size. At $N \geq 192$, the phase system achieves perfect discrimination and maximum mutual information (2.000 bits). Binary Hopfield drops to literally zero MI at $N \geq 128$. However, this scaling result should be interpreted cautiously: the number of families (3) and members per family (4) remain fixed while N increases, so the effective load α decreases. Perfect discrimination at large N may partly reflect concentration-of-measure effects in high-dimensional spaces rather than a general scaling law. Scaling N , family count, and family size simultaneously would provide a stronger test.

3.10 Experiment 9: Load Robustness

Question: Does discrimination survive at high pattern load?

$N=80$, 3 families $\times$ 4 members (85% similar), with increasing numbers of unrelated patterns. Note: this tests robustness to increasing network load, not classical Hopfield capacity (which measures

exact recall of all stored patterns). The measured objective is identity discrimination within the family subset.

Total patterns	α	Ph disc	Bn disc	Ph MI
12	0.150	93.9%	25.6%	1.804
17	0.212	93.3%	26.7%	1.781
22	0.275	93.3%	25.6%	1.796
32	0.400	91.1%	27.8%	1.731
42	0.525	92.8%	25.6%	1.764
52	0.650	96.1%	26.7%	1.874

Discrimination remains above 91% and MI remains above 1.73 bits even at $\alpha = 0.65$ (52 patterns in 80 neurons). The phase system's ability to discriminate family members is essentially insensitive to the number of unrelated patterns stored in the network.

3.11 Experiment 10: Discrimination vs Recall Across Similarity

Question: Are discrimination and exact recall measuring different things?

Similarity	Ph disc	Ph exact	Bn disc	Bn exact	Ph MI
70%	98.3%	74.6%	37.2%	77.5%	1.946
80%	96.1%	74.7%	25.6%	85.4%	1.874
85%	92.8%	74.7%	26.1%	87.7%	1.764
90%	88.3%	74.9%	25.6%	92.9%	1.646
95%	72.8%	75.4%	25.6%	95.9%	1.266

At 95% similarity, Hopfield achieves 95.9% exact recall versus the phase system's 75.4%. Yet Hopfield discriminates at chance (25.6%) while the phase system discriminates at 72.8%. Higher similarity makes Hopfield look better on bit-level metrics (because the prototype IS 95% similar to every member) while the discrimination task becomes harder. The two metrics measure fundamentally different properties of the retrieved representation.

3.12 Experiment 11: Resistance to Topological Sparsification (N=1024)

Question: Does the discrimination advantage survive sparse connectivity?

All previous experiments used dense (fully connected) weight matrices. Biological neural circuits are sparse — each neuron connects to a small fraction of the network. We tested all five coupling models under a directed random graph with fixed out-degree $k=256$ at $N=1024$ (25% connectivity). Synaptic weights on surviving edges were computed exactly from stored patterns using the Hebbian rule. Self-couplings were removed. 3 families $\times$ 4 members (85% similar), 64 unrelated patterns (76 total, $\alpha = 0.074$), 20 seeds.

Model	Coupling	Discrimination	$\pm$ Std
A. Sinusoidal	$\sin(\varphi_j - \varphi_i)$	100.0%	$\pm 0.0\%$
B. Linear circular	$(\varphi_j - \varphi_i) \bmod 2\pi$	100.0%	$\pm 0.0\%$
C. Cosine	$\cos(\varphi_j - \varphi_i)$	100.0%	$\pm 0.0\%$
D. tanh on circle	$\tanh(J @ \varphi)$	25.0%	$\pm 43.3\%$

E. Binary Hopfield	$\text{sign}(Wx)$	25.0%	$\pm 43.3\%$
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All three pairwise-difference models maintain perfect discrimination (100%) even with 75% of network connections removed. Both field-based models collapse to exactly $25.0\% \pm 43.3\%$ — matching the theoretical Bernoulli standard deviation $\sqrt{(0.25 \times 0.75)} = 0.433$ for random chance among 4 members. This confirms complete prototype collapse: the field models' outputs are statistically indistinguishable from random guessing.

The sparsity result is important for two reasons. First, it rules out the possibility that the discrimination advantage depends on global connectivity — pairwise coupling works with only local connections. Second, the biological plausibility of the mechanism increases, since cortical networks operate with sparse, predominantly local connectivity.

Note: The initial sparse implementation (generated by Copilot) contained a critical bug — it used mean-field approximation for edge weights instead of exact per-edge Hebbian computation, and the field models inadvertently bypassed the sparse topology. This was identified by Gemini's code review. The results reported here use the corrected implementation with exact sparse Hebbian weights and enforced topology for all models.

4. The Mechanism: Why Pairwise Phase-Difference Coupling Works

The discrimination advantage arises from a fundamental difference in how information propagates through each network type during retrieval.

Field-based coupling averages away fine-grained differences.

In both binary and continuous Hopfield, the update for neuron i depends on a global field: $h_i = \sum_j W_{ij} s_j$ (or $\tanh(\beta h_i)$). This field is a weighted sum over all neurons. When stored patterns are 85% similar, their contributions to the field are nearly identical. The field points toward the family centroid — the prototype — not toward any specific member. The fine-grained differences (which 15% of neurons distinguish this member from others) are lost in the sum.

Pairwise phase-difference coupling preserves local geometric relationships.

In the phase-coupled system, the update for oscillator i depends on pairwise differences: $d\phi_i = \eta \sum_j J_{ij} f(\phi_j - \phi_i)$. Although the sum runs over all connected neurons j , each term $f(\phi_j - \phi_i)$ is a function of the difference between two specific oscillators' phases, not of their absolute values. When family members differ in neuron k but agree in neuron m , the pairwise coupling between k and m carries information about which specific member the current configuration matches. Because each term depends on the relative geometry of oscillator pairs rather than on a global aggregate of individual states, the fine-grained angular differences between similar patterns propagate coherently through the network rather than being averaged away.

The coupling function appears secondary; the coupling structure is essential.

All three tested functions of the pairwise difference $(\phi_j - \phi_i)$ — sin, linear, cosine — produce identical discrimination (~93%). The essential structural property, within the tested scope, is that coupling depends on the difference between individual oscillator pairs, not on a global summation.

The circular topology enables but does not cause the effect. Model D ($\tanh$ on S^1) lives on the circle but uses field-based coupling. It fails. The circle provides the manifold on which phase differences are defined, but without pairwise coupling the topology alone conveys no advantage.

5. Discussion

5.1 Relationship to Prior Work

The observation that correlated patterns create interference in Hopfield networks is classical (Amit, 1989). Various solutions have been proposed: pseudoinverse learning rules (Personnaz, Guyon & Dreyfus, 1986), unlearning (Hopfield, Feinstein & Palmer, 1983), and modern dense associative memories (Krotov & Hopfield, 2016) that achieve exponential capacity through higher-order interactions. However, these approaches focus on capacity and storage, not on discrimination of similar patterns within families.

Oscillatory associative memories have been studied extensively (Aonishi, 1998; Nishikawa, Lai & Hoppensteadt, 2004). The Kuramoto model with learned couplings is well-established. Phase-coded memory in biological neural circuits is supported by substantial evidence (O'Keefe & Recce, 1993; Lisman & Jensen, 2013). However, to our knowledge, no prior work has explicitly demonstrated and ablated the family-discrimination advantage of pairwise phase-difference coupling over binary and continuous Hopfield baselines.

5.2 Implications for Associative Memory Theory

The finding suggests that the capacity-storage view of associative memory is incomplete. Capacity measures how many patterns can be stored and retrieved. Discrimination measures how finely the system can distinguish similar patterns. These are different properties, and a system can have moderate capacity but excellent discrimination (as the phase-coupled system shows) or high capacity but no discrimination (as modern Hopfield variants may show for correlated patterns).

The representational consequence is that Hopfield-type weight matrices encode pattern identity but not pattern similarity structure. The weight matrix $W = (1/N) \sum \xi_{\mu} \xi_{\mu}^T$ treats patterns as points; the space between points carries no retrievable information. Phase-coupled dynamics, by contrast, encode patterns as configurations on a continuous manifold where the geometry between patterns is accessible to the retrieval dynamics.

5.3 Implications for Biological Memory

Phase-coded memory in hippocampal and cortical circuits (phase precession, theta-gamma coupling, oscillatory population codes) is a central feature of biological neural computation. The present finding suggests a computational rationale beyond timing and sequencing: phase-coupled dynamics may enable biological circuits to discriminate between highly similar memories that rate-coded (binary or continuous) representations would confound.

This is speculative and no biological data are presented. However, the prediction is specific: brain regions using phase-coded representations should show superior discrimination between similar stimuli compared to regions using rate-coded representations, as measured by decoding accuracy from population recordings.

6. Limitations

- Scale: Dense experiments use $N = 80$ with 17 patterns ($\alpha = 0.212$). Scaling was tested at $N = 128, 192, 256$ (dense) and $N = 1024$ (sparse, 25% connectivity) with fixed family structure. In all cases the family count (3) and member count (4) remained fixed while N increased, so the effective load decreases with N . Simultaneous scaling of N , family count, and family size has not been tested and is needed before claiming a general scaling law.

- Family structure: Families are generated by flipping random bits in a prototype. This creates a specific geometric structure. Other correlation structures (hierarchical, manifold-embedded, semantic) have not been tested.
- Family size: Only 4 members per family tested. Larger families (8, 16, 32) may reveal capacity limits in the discrimination mechanism.
- Modern baselines: We compare against standard Hopfield and continuous Hopfield. Dense associative memories (Ramsauer et al., 2020), modern Hopfield networks (Krotov & Hopfield, 2016), and other advanced architectures have not been tested.
- Retrieval metric: Discrimination accuracy uses an external nearest-neighbor classifier on the binarized output. The network itself does not explicitly output a family-member label. The metric measures whether the information exists in the output, not whether the network explicitly computes it.
- Jitter asymmetry: Phase systems receive small Gaussian jitter ($\sigma = 0.15$) to activate dynamics, which is absent from the binary Hopfield baseline. This is probably not responsible for a 66.7% gap but has not been formally excluded.
- Theoretical understanding: We identify pairwise phase-difference coupling as the critical ingredient but do not provide a formal mathematical proof of why it preserves family-member information. The mechanism description in Section 4 is informal.
- Biological relevance: All claims about biological memory are speculative. No neural data are presented.
- Task definition: This study measures discrimination (identity preservation within similarity clusters), not classical associative memory recall. We do not claim superior memory retrieval capacity. The phase system achieves lower exact bit-level recall (74.7%) than Hopfield (87.7%) while preserving more member-specific information. These are different computational objectives.
- MI measurement: The mutual information reported is $I(\text{target member}; \text{nearest-neighbor prediction})$, not $I(\text{target member}; \text{raw network state})$. The nearest-neighbor decoder is a legitimate classifier but introduces structure into the measurement. A held-out trained decoder would provide a more rigorous information-theoretic estimate.
- Scaling interpretation: Perfect discrimination at $N \geq 192$ occurs while family count and member count remain fixed. The effective load α decreases as N grows. This may reflect concentration-of-measure effects rather than a general scaling law. Testing with simultaneously increasing N , family count, and family size would be more informative.

7. Conclusion

Through eleven progressive experiments, each designed to address the strongest objection to the previous result, we identify a specific and robust phenomenon: in synthetic correlated-family memory tasks, phase-coupled Hebbian dynamics preserve substantially more member-specific information than binary and continuous Hopfield baselines, despite lower exact bit-level recall. Under matched-noise and matched-evaluation conditions, the phase model achieves high family-member discrimination and mutual information while both Hopfield variants collapse toward family prototypes and preserve little member-specific information. The effect survives sparse connectivity (25% at $N=1024$), confirming it does not depend on dense network structure.

The effect is large (92.5% vs 25.8% discrimination; 1.764 vs 0.049 bits MI under matched conditions), statistically significant ($p < 0.0003$, 15–20 seeds), robust across similarity levels (70–95%), and survives comprehensive ablation: amplitude dynamics, coherence priors, computational budget, noise model, evaluation metric, coupling function, and continuous

baselines have all been controlled. The minimal surviving system is bare Hebbian-weighted pairwise phase-difference coupling — a minimal oscillatory associative memory.

The critical distinction is not between continuous and discrete, nor between circular and linear topology. It is between pairwise-difference coupling structure and global-field coupling structure. The tested coupling functions that depended on pairwise phase differences retained discrimination, whereas the tested field-based dynamics did not.

Perhaps the most scientifically interesting finding is the exact-recall inversion: Hopfield achieves higher bit-level accuracy by converging to the family prototype, while the phase system achieves lower bit-level accuracy but preserves which specific member generated the cue. This reveals a separation between identity preservation and bit-level reconstruction that standard recall metrics are blind to.

This finding does not establish a fundamentally superior memory principle or a general superiority over all Hopfield-family architectures. Modern dense associative memories, attention-style retrieval, and other advanced architectures remain untested. What the evidence supports is a specific, reproducible, and well-ablated benchmark result that identifies pairwise phase-difference coupling as a mechanism for preserving fine-grained pattern identity under high similarity — a property worth investigating further across broader families of architectures and correlation structures.

8. AI Contribution Statement

This work was conducted through sustained collaboration with multiple AI systems. All theoretical insights, experimental designs, and scientific judgments are the author's. AI contributions:

- Claude (Anthropic): Primary computational collaborator. Designed and implemented experiments 1–10, performed ablations, wrote analysis code, and co-authored this manuscript. Claude also identified the initial similarity-discrimination hypothesis and proposed the mechanism-first experimental strategy.
- ChatGPT (OpenAI): Primary critical reviewer. Identified the noise-model asymmetry, requested the continuous Hopfield baseline, proposed the component ablation and phase decomposition experiments, and repeatedly challenged overclaiming. The OMA ablation — which destroyed a previous architectural hypothesis — was prompted by ChatGPT's identification of the $\max(a_1, a_2)$ confound.
- Gemini (Google): Proposed the DMS architecture (which was found to contain a fatal oracle flaw upon code review). Identified critical bug in Copilot's sparse implementation (mean-field approximation instead of exact per-edge Hebbian weights). Contributed conceptual framing.
- Copilot (Microsoft): Generated initial sparse and low-rank scaling implementations. The sparse implementation contained a structural bug (mean-field weight approximation and topology bypass for field models) that was identified by Gemini and corrected before results were reported.
- Grok (xAI): Provided independent stochastic simulations for the GHOD avalanche statistics in earlier work.

The multi-AI review process — showing results from one system to another for critical evaluation — was essential for identifying confounds and preventing overclaiming. The OMA ablation and the noise-model correction are direct products of this process.

9. Code Availability

All code is publicly available at <https://github.com/labinot-marku/CONN-Associative-Memory/tree/main/similarity-discrimination>. Six experiment scripts, all using only NumPy and SciPy, all deterministic with fixed random seeds:

- `conn_similarity.py` — Initial discovery (Experiment 1)
- `conn_similarity_ablation.py` — Component ablation + similarity sweep + computational budget (Experiments 2–4)
- `conn_fair_comparison.py` — Matched conditions + continuous Hopfield baseline (Experiment 5)
- `conn_phase_decomposition.py` — Phase system decomposition (Experiments 6–7)
- `conn_final_validation.py` — Mutual information, scaling, capacity load, discrimination vs recall (Experiments 7–10)
- `conn_sparse_validation.py` — Sparse connectivity test at $N=1024$ with exact per-edge Hebbian weights (Experiment 11)

All results have been independently reproduced in Google Colab with identical numbers.

10. References

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